Posts, Two Ways

Name: _____ Date: _____

Records or an abstract class?

Two versions of the same three kinds of post. Both start with `interface Post { String preview(); }`.

Version R: records

```
record TextPost(String user, String message,
    boolean edited) implements Post {
  public String preview() {
    if (edited) {
      return "@" + user + ": " + message
          + " (edited)";
    }
    return "@" + user + ": " + message;
  }
}
record VideoPost(String user, String message,
    int seconds) implements Post {
  public String preview() {
    return "@" + user + ": " + message
        + " [" + seconds + "s]";
  }
}
record Repost(String user, String message,
    String from) implements Post {
  public String preview() {
    return "@" + user + " reposted @" + from
        + ": " + message;
  }
}
```

Version A: an abstract class

```
abstract class Message implements Post {
  String user;
  String message;
  Message(String user, String message) {
    this.user = user;
    this.message = message;
  }
}
class TextPost extends Message {
  boolean edited;
  TextPost(String user, String message,
      boolean edited) {
    super(user, message);
    this.edited = edited;
  }
  public String preview() { /* same as R */ }
}
class VideoPost extends Message {
  int seconds;
  VideoPost(String user, String message,
      int seconds) {
    super(user, message);
    this.seconds = seconds;
  }
  public String preview() { /* same as R */ }
}
class Repost extends Message {
  String from;
  Repost(String user, String message,
      String from) {
    super(user, message);
    this.from = from;
  }
  public String preview() { /* same as R */ }
}
```

Q1. Before counting: which version would you rather have written? Why?

Q2. Written out in full, with each `preview` as in R and one statement per line, R is 17 lines and A is 40. Where do A's extra lines go?

Q3. Circle what this prints in each version, and say why they differ.

```
System.out.println(new TextPost("ana", "hi", false).equals(new TextPost("ana", "hi", false)));
```

R: true / false A: true / false

Q4. Every post now needs a `String date`, given when the post is made. In each version, which lines change? Predict which version needs fewer edits, then count.

R: A:

Q5. Posts now get `int likes`, starting at 0, and a method `void like()` that adds one. What has to change in each version?

R: A:

One class per kind, or one class with a `kind`?

Version K: a class per kind

```
interface Post {
  String preview();
  boolean isLong();
}
record TextPost(String user, String message)
    implements Post {
  public String preview() {
    return "@" + user + ": " + message;
  }
  public boolean isLong() {
    return message.length() > 80;
  }
}
record VideoPost(String user, String message,
    int seconds) implements Post {
  public String preview() {
    return "@" + user + ": " + message
        + " [" + seconds + "s]";
  }
  public boolean isLong() { return seconds > 60; }
}

Post a = new TextPost("ana", "Exam moved to Friday");
Post b = new VideoPost("ben", "My cat learns Java",
42);
```

Version F: one record with a `kind` field

```
record Post(String kind, String user,
    String message, int seconds) {
  String preview() {
    if (kind.equals("video")) {
      return "@" + user + ": " + message
          + " [" + seconds + "s]";
    }
    return "@" + user + ": " + message;
  }
  boolean isLong() {
    if (kind.equals("video")) {
      return seconds > 60;
    }
    return message.length() > 80;
  }
}

Post a = new Post("text", "ana",
    "Exam moved to Friday", 0);
Post b = new Post("video", "ben",
    "My cat learns Java", 42);
```

Q6. In F, what does the `0` in `a` mean? What does `new Post("vidoe", "ben", "My cat learns Java", 42).preview()` return (note `"vidoe"`, a typo for `"video"`), and what would the same typo do in K?

Q7. Make each change in both versions, and count the places you touch. Count a new class, a new method, an edited line, or an edited `new` as one place each.

change	K: places	F: places
Add link posts: a `String url`, previewed as `"@ana: Read this <url>"`, never long.		
Add `boolean mentions(String word)`: true when the message contains `word`.		

Q8. A feed app adds a new kind of post every month, but rarely a new operation. Which version would you pick? Which if it is the other way around?